

Data Exchange Framework

DEF Doctor



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DEF Doctor

DEF Doctor is a command line tool designed to diagnose and identify Data Exchange Framework system and configuration properties/settings issues independently of the DEF. It is delivered as a single self-contained runnable JAR file. DEF Doctor can be directed to a specific DEF instance or can be run in “auto-discovery” mode. The DEF Doctor auto-discovery feature searches all drives and all folders, apart from the Microsoft Windows folder, for installed DEF systems. Auto-discovery looks specifically for `ndef.properties` files.

Important

DEF Doctor is designed to work with 3.x, 4.x and 5.x versions of the DEF.

The tool performs the following checks:

- System checks: Twenty three different environment/system checks, including checks for available memory, disk space, database connectivity, file and folder access, service configuration, workbench installation, etc.
- Configuration checks: Scans all configured customers, sites, server, processes, process components and resources for a specific installed and configured DEF instance. Checks include file and SFTP communications folder access, process component checks, etc.

Prerequisites

- An installed Java JRE (either DEF JRE or external JRE).

COMMAND LINE OPTIONS

All output is to the command window. The DEF Doctor supports the following command line parameters:

OPTION	DESCRIPTION
-showChecks	Show all available DEF Doctor configuration checks
-verbose	Log all messages (INFO, SUCCESS, WARNING, FAILURE) (Default is NOT verbose)
-color	Show console output in color. Default is NO color. If the -color command line argument is specified, then the following colors are used to render the console window output: • Blue - Informational • Green - Success • Yellow - Warning • Red - Error Color console output is NOT supported on all Windows systems. If color is not supported, then do not use the -color parameter NOTE: To view color console output, the ansicon driver may need to be installed. To install the ansicon driver, run the following command from the command window prompt: <pre>ansicon -i</pre> Use this link to download and run asicon: http://qastuffs.blogspot.com/2011/02/how-to-install-ansicon-for-cucumber-to.html
-path	Optional path to a specified ndef.properties file, if a path is not specified, then the DEF Doctor will auto discover all installed DEF systems on the current machine.
-systemChecksOnly	Perform system checks only
-configurationChecks Only	Perform configuration checks only

RUNNING THE DEF DOCTOR JAR FILE

The DEF Doctor is delivered as a single jar file. For the most accurate results, run the DEF Doctor using the same service account user that the DEF service runs under.

- Use the DOS "runas" command to open a command window using the DEF service account user. For example:

```
runas /profile /env /user:fast-talk\gal "cmd"
```

Example 1

- Run the following command from the service account command window:

```
"C:\Program Files\Java\jdk1.8.0_171\bin\java" -jar DEFDoctor.jar -verbose -color -  
path="C:\Users\cowen\Perforce\cowen-def-rel-5.0\def\NEAT\etc\ndef.properties"
```

- Must supply the full path to a JAVA JRE/JDK. This can be either the JRE that is embedded with the DEF or an external JRE/JDK.
- The Java library path (-Djava.library.path) does NOT need to be specified.
- The DEF Doctor assumes the Java library path is .\Data Exchange Framework\data\workbench\lib.
- Must specify the name of the runnable jar file: -jar DEFDoctor.jar.
- This example runs with full console logging. If -verbose is NOT specified, then only warning and error messages are written to the console window.
- This example attempts to render the console output in color. ansicon.exe is required to view console output in color.
- This example specifies a -path parameter to a specific ndef.properties file. When a -path parameter is specified, then auto discovery will not occur.
- The DEF Doctor can be run from ANY location on a specific machine. The DEF Doctor does NOT have to be run from the location where a DEF system is installed.

Example 2

```
"C:\Program Files\Java\jdk1.8.0_171\bin\java" -jar DEFDoctor.jar -verbose -color
```

- This example does NOT specify a -path parameter. In this case, the DEF Doctor attempts to auto-discover all DEF systems installed on the current machine.

SYSTEM AND CONFIGURATION CHECKS

After running the JAR file, the following checks are done:

System Checks

System health checks are run based on `ndef.properties` and system in general:

- Check for Java library path
- Check for `ndef.properties` file
- Check DEF version number
- Check DEF server type
- Check the DEF installation folder
- Check the DEF current user
- Check the available amount of memory
- Check the available number of cores
- Check the available disk size
- Check for previous DEF installations
- Check the DEF data folder for write access
- Check the DEF Compute Node data folder for write access
- Check for DEF installation folder for write access
- Check the DEF database connection to the DEF NEAT/NEST database
- Check the SQL Server version number
- Check if the DEF NEAT/NEST database exists
- Check if the DEF NEAT/NEST database has been initialized
- Check if the Java Cryptography Extension is installed
- Check if the Nexidia Workbench is installed
- Check if the Nexidia Workbench license has expired
- Check if the configured DEF service is running
- Check the DEF service owner
- Check the DEF system log level

Configuration Checks

These are used to check company site, server, process, resource configurations. After these checks, the system retrieves configuration hierarchy and scans all hierarchy elements.

- Check the customer configuration
- Check the site configuration
- Check the site file system communications folder
- Check the site SFTP communications folder
- Check the server configuration
- Check the server file system communications folder
- Check the server SFTP communications folder
- Check the server log level
- Check the server communications folder uniqueness
- Check the process configuration
- Check the process component key store configuration
- Check the process component crypto database connection configuration
- Check the process component decrypt stored procedure configuration
- Check the process component Rabbit MQ configuration
- Check the file system resource configuration
- Check the SFTP resource configuration
- Check the database resource configuration